Université de Genève

Départment d'informatique

Éléments de la Théorie de l'information

Information Theory for Data Science and Machine Learning — 18.03.2020

Problem Set 4 — Due: 25.03.2020

Note: Please follow the instruction on `http://sip.unige.ch/education/information-theory/`.

Problem 1 (Markov Process and EVD—25 points).
Please consider the lecture notes Slide 98 (Theme 1). Obtain matrices $\mathbf{U}$ and $\mathbf{D}$ for $\sigma_X^2 = 1$, $N = 4$, $\rho = 0.1, 0.5, 0.95$. Make conclusions about eigenvalue matrix properties.

$$\mathbf{K}_{\mathbf{xx}} = \sigma_X^2 \begin{bmatrix} 1 & \rho & \rho^2 & \rho^3 \\ \rho & 1 & \rho & \rho^2 \\ \rho^2 & \rho & 1 & \rho \\ \rho^3 & \rho^2 & \rho & 1 \end{bmatrix} \quad N = 4 \quad \rho = 0.95$$

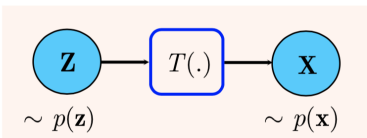
```
U =                                                  Low frequency
  High frequency
   0.2747   -0.5062    0.6516    0.4937
  -0.6516    0.4937    0.6516    0.5062
   0.6516    0.4937   -0.2747    0.5062
  -0.2747   -0.5062   -0.6516    0.4937

D =
   0.0300         0         0         0
        0    0.0506         0         0
        0         0    0.1627         0
        0         0         0    3.7568
```

Hint: You can use MATLAB/Python functions for eigenvalue decomposition (EVD).

Problem 2 (Explaining a few concepts—25 points).

- Explain the link between eigen-value and singular-value decompositions.
- Explain for which matrices we can apply each decomposition.
- Can we apply the SVD for the decomposition of covariance matrices? What one will obtain?

Problem 3 (Prove—25 points).
Let consider $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(\bar{\mathbf{z}}, \mathbf{K}_{\mathbf{zz}})$. Please consider the lecture notes Slide 99 (Theme 1). Try to re-prove the derivations provided there and obtain $p(\mathbf{x})$ as depicted in the Figure below for the above $\mathbf{z}$.



$$\mathbf{x} = \underbrace{\mathbf{W}^T}_{\mathbf{A}} \mathbf{z} = \mathbf{A}\mathbf{z}$$
$$\mathbf{z} = \mathbf{A}^{-1}\mathbf{x}$$

$$p_{\mathbf{x}}(\mathbf{x}) = \frac{p_{\mathbf{z}}\left(T^{-1}(\mathbf{x})\right)}{\left|\det J_T(\mathbf{z})\right|} = \frac{p_{\mathbf{z}}\left(\mathbf{A}^{-1}\mathbf{x}\right)}{\left|\det J_T(\mathbf{z})\right|} = \frac{p_{\mathbf{z}}\left(\mathbf{A}^{-1}\mathbf{x}\right)}{\left|\det \mathbf{A}\right|}$$

Make your conclusions about:

- What is the impact of a linear (affine) transform on:

  (a) mean

  (b) covariance matrix

  of input random vector.

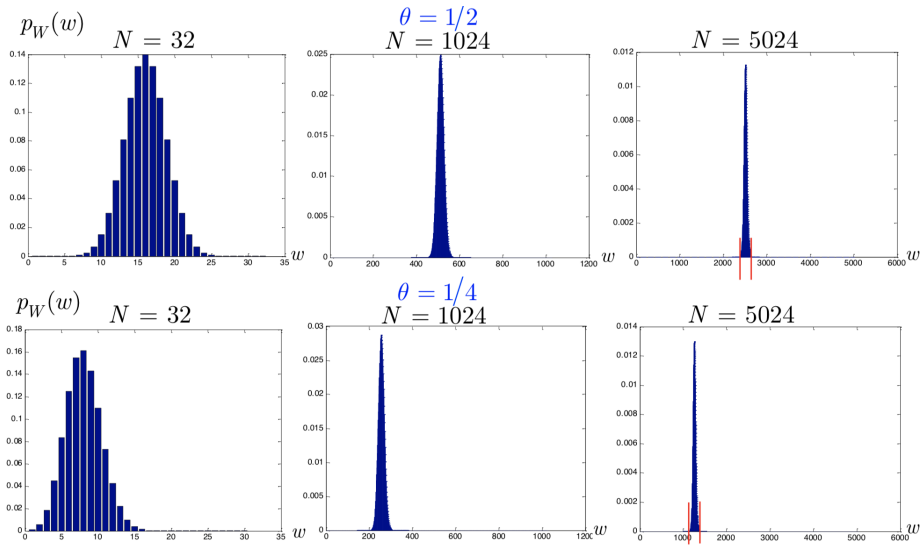
- Find:

  (a) mean and covariance matrix directly

  (b) pdf $p_{\mathbf{x}}(\mathbf{x})$ as above

- Compare your results

Problem 4 (To simulate Hamming weight and see the concentration properties in action—25 points).
Please consider the lecture notes Slides 111 to 117 (Theme 1) Plot the Hamming weight for $\theta = 0.25, 0.5$ and $N = 32, 1024, 5024$ and obtain the figures depicted in Slide 117.



Remark: Use simulation to obtain the above plots. On the top of them add the Binomial pmf's with the analytically found parameters. Compare these results and make conclusions.